

Total No. of Questions : 10]

SEAT No. :

P6523

[Total No. of Pages : 2

[5870]-2004

T.E. (Computer Engineering)

DIGITAL SIGNAL PROCESSING APPLICATIONS

(2012 Pattern) (Semester - II) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

Q1) a) Define mean, standard deviation, SNR, histogram and probability density function. [5]

b) State and explain sampling theorem and aliasing effect. [5]

OR

Q2) a) Find the circular convolution of $x(n) = \{1, 2, 2, 1\}$ and $h(n) = \{1, 2, 3, 1\}$ using any one method. [5]

b) State and define the periodic and non-periodic signal, even and odd signal also energy and power signal. [5]

Q3) a) Explain the Radix-2 DIT FFT algorithms. [5]

b) Obtain the Z Transform of sequence $x(n) = a^n u(n)$ and sketch the ROC. [5]

OR

Q4) a) State and explain any three properties in terms of Fourier transform. [5]

b) Describe the characteristic of filter and all pass filters. [5]

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- Q5)** a) Explain the Radix-2 DIF FFT algorithms and consider $N = 8$. [9]
b) State and explain causality, stability and initial value theorem in terms of z-transform. [9]

OR

- Q6)** a) What are filter structures? Explain the any application filters. [9]
b) Derive the Direct Form - II IIR filter structure from system function $H(Z)$ and represent it using multipliers, adders and delay elements. [9]

- Q7)** a) Explain important components of ADSP 21XXX family. [8]
b) Explain the application of DSP in telecommunication and biomedical. [8]

OR

- Q8)** a) Explain the features of SHARC DSP processor. List the number of DAGs with its capabilities and memory pointer registers supported by DAG. [8]
b) Write short note on OMAP multimedia applications. [8]

- Q9)** a) Explain in brief the digital image processing. Write short note on image enhancement. [8]
b) Explain the important components of DSPs. [8]

OR

- Q10)** a) Explain the high fidelity audio in brief and also explain limitation of DSP. [8]
b) Explain the brightness, contrast adjustment and gray scale transformation in brief. [8]

